

- 1-2 Verify that ∇T and $\nabla \cdot \mathbf{q}$ in the spherical coordinate system (r, ϕ, θ) are given as

$$\nabla T = \hat{\mathbf{u}}_r \frac{\partial T}{\partial r} + \hat{\mathbf{u}}_\phi \frac{1}{r \sin \theta} \frac{\partial T}{\partial \phi} + \hat{\mathbf{u}}_\theta \frac{1}{r} \frac{\partial T}{\partial \theta}$$

$$\nabla \cdot \mathbf{q} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 q_r) + \frac{1}{r \sin \theta} \frac{\partial q_\phi}{\partial \phi} + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (q_\theta \sin \theta)$$

- 1-12 Set up the mathematical formulation of the following heat conduction problems:

1. A slab in $0 \leq x \leq L$ is initially at a temperature $F(x)$. For times $t > 0$, the boundary at $x = 0$ is kept insulated and the boundary at $x = L$ dissipates heat by convection into a medium at zero temperature.
2. A semiinfinite region $0 \leq x < \infty$ is initially at a temperature $F(x)$. For times $t > 0$, heat is generated in the medium at a constant rate of $g_0 \text{ W/m}^3$, while the boundary at $x = 0$ is kept at zero temperature.
3. A solid cylinder $0 \leq r \leq b$ is initially at a temperature $F(r)$. For times $t > 0$, heat is generated in the medium at a rate of $g(r)$, W/m^3 , while the boundary at $r = b$ dissipates heat by convection into a medium at zero temperature.
4. A solid sphere $0 \leq r \leq b$ is initially at temperature $F(r)$. For times $t > 0$, heat is generated in the medium at a rate of $g(r)$, W/m^3 , while the boundary at $r = b$ is kept at a uniform temperature T_0 .

- 1-15 A long cylindrical iron bar of diameter $D = 5 \text{ cm}$, initially at temperature $T_0 = 650^\circ\text{C}$, is exposed to an air stream at $T_\infty = 50^\circ\text{C}$. The heat transfer coefficient between the air stream and the surface of the bar is $h = 80 \text{ W/(m}^2 \cdot ^\circ\text{C)}$. Thermophysical properties may be taken as $\rho = 7800 \text{ kg/m}^3$, $C_p = 460 \text{ J/(kg} \cdot ^\circ\text{C)}$, and $k = 60 \text{ W/(m} \cdot ^\circ\text{C)}$. Determine the time required for the temperature of the bar to reach 250°C by using the lumped system analysis.

- 1-16 A thermocouple is to be used to measure the temperature in a gas stream. The junction may be approximated as a sphere having thermal conductivity $k = 25 \text{ W/(m} \cdot ^\circ\text{C)}$, $\rho = 8400 \text{ kg/m}^3$, and $C_p = 0.4 \text{ kJ/(kg} \cdot ^\circ\text{C)}$. The heat transfer coefficient between the junction and the gas stream is $h = 560 \text{ W/(m}^2 \cdot ^\circ\text{C)}$. Calculate the diameter of the junction if the thermocouple should measure 95% of the applied temperature difference in 3s.

- 2-2 A slab, $0 \leq x \leq L$, is initially at a temperature $F(x)$. For times $t > 0$ the boundary surface at $x = 0$ is kept insulated and that at $x = L$ dissipates heat by convection into a medium at zero temperature with a heat transfer coefficient h . Obtain an expression for the temperature distribution $T(x, t)$ in the slab for times $t > 0$ and for the heat flux at the boundary surface $x = L$.

- 2-6 In a one-dimensional infinite medium, $-\infty < x < \infty$, initially, the region $a < x < b$ is at a constant temperature T_0 , and everywhere outside this region is at zero temperature. Obtain an expression for the temperature distribution $T(x, t)$ in the medium for times $t > 0$.

- 2-15 Obtain an expression for the steady-state temperature distribution $T(x, y)$ in a rectangular region $0 \leq x \leq a, 0 \leq y \leq b$ for the following boundary conditions: the boundary at $x = 0$ is kept insulated, the boundary at $y = 0$ is kept at a temperature $f(x)$ and the boundaries at $x = a$ and $y = b$ dissipate heat by convection into an environment at zero temperature. Assume the heat transfer coefficient to be the same for both boundaries.

Obtain an expression for the steady-state temperature distribution $T(x, y, z)$

1-2 Verify that ∇T and $\nabla \cdot \mathbf{q}$ in the spherical coordinate system (r, ϕ, θ) are given as

$$\nabla T = \hat{u}_r \frac{\partial T}{\partial r} + \hat{u}_\phi \frac{1}{r \sin \theta} \frac{\partial T}{\partial \phi} + \hat{u}_\theta \frac{1}{r} \frac{\partial T}{\partial \theta}$$

$$\nabla \cdot \mathbf{q} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 q_r) + \frac{1}{r \sin \theta} \frac{\partial q_\phi}{\partial \phi} + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (q_\theta \sin \theta)$$

From Eqs. (1-19) and (1-22) we have

$$\nabla T = \hat{u}_1 \frac{1}{a_1} \frac{\partial T}{\partial u_1} + \hat{u}_2 \frac{1}{a_2} \frac{\partial T}{\partial u_2} + \hat{u}_3 \frac{1}{a_3} \frac{\partial T}{\partial u_3} \quad (1-19)$$

$$\nabla \cdot \mathbf{q} = \frac{1}{a} \left[\frac{\partial}{\partial u_1} \left(\frac{a}{a_1} q_1 \right) + \frac{\partial}{\partial u_2} \left(\frac{a}{a_2} q_2 \right) + \frac{\partial}{\partial u_3} \left(\frac{a}{a_3} q_3 \right) \right] \quad (1-22a)$$

$$\text{where } a = a_1 a_2 a_3 \quad (1-22b)$$

The relationship between (r, ϕ, θ) and (x, y, z) are

$$x = r \sin \theta \cos \phi, \quad y = r \sin \theta \sin \phi, \quad z = r \cos \theta$$

$$\text{Let } u_1 = r, \quad u_2 = \phi \quad \text{and} \quad u_3 = \theta$$

$$\text{From (1-18): } a_i^2 = \left(\frac{\partial x}{\partial u_i} \right)^2 + \left(\frac{\partial y}{\partial u_i} \right)^2 + \left(\frac{\partial z}{\partial u_i} \right)^2$$

$$a_1 = a_r = 1, \quad a_2 = a_\phi = r \sin \theta, \quad a_3 = a_\theta = r$$

Then Eq. (1-19) gives

$$\nabla T = \hat{u}_r \frac{\partial T}{\partial r} + \hat{u}_\phi \frac{1}{r \sin \theta} \frac{\partial T}{\partial \phi} + \hat{u}_\theta \frac{1}{r} \frac{\partial T}{\partial \theta}$$

and Eq. (1-22a) yields

$$\nabla \cdot \mathbf{q} = \frac{1}{r^2 \sin \theta} \left[\frac{\partial}{\partial r} (r^2 \sin \theta q_r) + \frac{\partial}{\partial \phi} \left(\frac{r^2 \sin \theta}{r} q_\phi \right) + \frac{\partial}{\partial \theta} \left(\frac{r^2 \sin \theta}{r} q_\theta \right) \right]$$

or

$$\nabla \cdot \mathbf{q} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 q_r) + \frac{1}{r^2 \sin \theta} \frac{\partial q_\phi}{\partial \phi} + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta q_\theta)$$

1-12 Set up the mathematical formulation of the following heat conduction problems:

1. A slab in $0 \leq x \leq L$ is initially at a temperature $F(x)$. For times $t > 0$, the boundary at $x = 0$ is kept insulated and the boundary at $x = L$ dissipates heat by convection into a medium at zero temperature.
2. A semi-infinite region $0 \leq x < \infty$ is initially at a temperature $F(x)$. For times $t > 0$, heat is generated in the medium at a constant rate of g_0 W/m³, while the boundary at $x = 0$ is kept at zero temperature.
3. A solid cylinder $0 \leq r \leq b$ is initially at a temperature $F(r)$. For times $t > 0$, heat is generated in the medium at a rate of $g(r)$ W/m³, while the boundary at $r = b$ dissipates heat by convection into a medium at zero temperature.
4. A solid sphere $0 \leq r \leq b$ is initially at temperature $F(r)$. For times $t > 0$, heat is generated in the medium at a rate of $g(r)$ W/m³, while the boundary at $r = b$ is kept at a uniform temperature T_0 .

$$1. \frac{\partial^2 T}{\partial x^2} = \frac{1}{\alpha} \frac{\partial T}{\partial t} \quad \text{in } 0 < x < L, \quad t > 0$$

$$\frac{\partial T}{\partial x} = 0 \quad \text{at } x = 0, \quad t > 0$$

$$-k \frac{\partial T}{\partial x} + hT = 0 \quad \text{at } x = L, \quad t > 0$$

$$T = F(x) \quad \text{for } t = 0, \quad 0 \leq x \leq L$$

$$2. \frac{\partial^2 T}{\partial x^2} + \frac{1}{k} g_0 = \frac{1}{\alpha} \frac{\partial T}{\partial t} \quad \text{in } 0 < x < \infty, \quad t > 0$$

$$T = 0 \quad \text{at } x = 0, \quad t > 0$$

$$T = F(x) \quad \text{for } t = 0, \quad 0 \leq x < \infty$$

$$3. \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial T}{\partial r} \right) + \frac{1}{k} g(r) = \frac{1}{\alpha} \frac{\partial T}{\partial t} \quad \text{in } 0 \leq r < b, \quad t > 0$$

$$\frac{\partial T}{\partial r} + HT = 0 \quad \text{at } r = b, \quad t > 0$$

$$T = F(r) \quad \text{for } t = 0, \quad 0 \leq r < b$$

$$\frac{\partial T}{\partial r} = 0 \quad \text{at } r = 0, \quad \text{for } t \geq 0$$

$$4. \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial T}{\partial r} \right) + \frac{1}{k} g = \frac{1}{\alpha} \frac{\partial T}{\partial t} \quad \text{in } 0 \leq r$$

$$T = T_0 \quad \text{at } r = b, \quad t > 0$$

$$\frac{\partial T}{\partial r} = 0 \quad \text{at } r = 0, \quad t > 0$$

$$T = F(r) \quad \text{for } t = 0, \quad 0 \leq r < b$$

1-15 Along cylindrical iron bar of diameter $D = 5$ cm, initially at temperature $T_0 = 650^\circ\text{C}$ is exposed to an air stream at $T_\infty = 50^\circ\text{C}$. Heat transfer coefficient between the air stream and the surface of the bar is $h = 80$ W/(m²·°C). Thermophysical properties may be taken as: $\rho = 7800$ kg/m³, $C_p = 460$ J/(kg·°C) and $k = 60$ W/(m·°C). Determine the time required for the temperature of the bar to reach 250°C by using the lumped system analysis.

From Eq. (1-69) we have

$$\frac{T(t) - T_\infty}{T_0 - T_\infty} = e^{-mt}$$

$$\text{where } m = \frac{hA}{\rho C_p V} = \frac{h}{\rho C_p} \left(\frac{2\pi r^2 + 2\pi rL}{\pi r^2 L} \right) \approx \frac{h}{\rho C_p} \frac{2\pi rL}{\pi r^2 L}$$

$$= \frac{h}{\rho C_p} \frac{2}{r} = \frac{80}{7800 \times 460} \frac{2}{0.025} = 1.784 \times 10^{-3} \text{ sec}^{-1}$$

$$\text{Then } \frac{250 - 50}{650 - 50} = \exp(-1.784 \times 10^{-3} t)$$

$$\text{or } t = 616 \text{ sec} = 10 \text{ min } 16 \text{ sec}$$

1-16 A thermocouple is to be used to measure the temperature in a gas stream. The junction may be approximated as a sphere having thermal conductivity $k = 25$ W/(m·°C), $\rho = 8400$ kg/m³ and $C_p = 0.4$ kJ/(kg·°C). The heat transfer coefficient between the junction and the gas stream is $h = 560$ W/(m²·°C). Calculate the diameter of the junction if the thermocouple should measure 95 percent of the applied temperature difference in 3s.

$$\frac{A}{V} = \frac{4\pi r^2}{(4/3)\pi r^3} = \frac{3}{r} = \frac{6}{D}$$

$$m = \frac{hA}{\rho C_p V} = \frac{560}{8400 \times 400} \frac{6}{D} = \frac{0.001}{D} \text{ sec}^{-1}$$

To measure 95% of the applied temperature difference in 3 sec, the junction diameter is determined as

$$\frac{5}{100} = e^{-mt} = e^{-\frac{0.001}{D} \times 3} \quad \therefore D = 10^{-3} \text{ m} = 1 \text{ mm}$$

2-2 A slab, $0 \leq x \leq L$ is initially at a temperature $F(x)$. For times $t > 0$ the boundary surface at $x = 0$ is kept insulated and that at $x = L$ dissipates heat by convection into a medium at zero temperature with a heat transfer coefficient h . Obtain an expression for the temperature distribution $T(x, t)$ in the slab for times $t > 0$ and for the heat flux at the boundary surface $x = L$.

The mathematical formulation of the problem is given as

$$\begin{aligned}\frac{\partial^2 T}{\partial x^2} &= \frac{1}{\alpha} \frac{\partial T}{\partial t} & \text{in } 0 < x < L, t > 0 \\ \frac{\partial T}{\partial x} &= 0 & \text{at } x = 0, t > 0 \\ \frac{\partial T}{\partial x} + HT &= 0 & \text{at } x = L, t > 0 \\ T &= F(x) & \text{for } t = 0 \text{ in } 0 \leq x \leq L.\end{aligned}$$

The formal solution is given by Eq (2-36) as

$$T(x, t) = \sum_{m=1}^{\infty} \tilde{e}^{-\alpha \beta_m^2 t} \frac{1}{N(\beta_m)} X(\beta_m, x) \int_0^L X(\beta_m, x') F(x') dx' \quad (1)$$

and from Table 2-2, Case # 4 we have

$$X(\beta_m, x) = \cos \beta_m x, \quad \frac{1}{N(\beta_m)} = 2 \frac{\beta_m^2 + H^2}{L(\beta_m^2 + H^2) + H} \quad (2a)$$

where β_m 's are the roots of

$$\beta_m \tan \beta_m L = H \quad (2b)$$

Substituting Eqs (2) into the formal solution (1) we obtain

$$T(x, t) = 2 \sum_{m=1}^{\infty} \tilde{e}^{-\alpha \beta_m^2 t} \frac{\beta_m^2 + H^2}{L(\beta_m^2 + H^2) + H} \cos \beta_m x \int_0^L \cos \beta_m x' F(x') dx' \quad (3)$$

The heat flux at the boundary surface $x = L$ is determined as

$$q = -k \frac{\partial T}{\partial x} \Big|_{x=L} = 2k \sum_{m=1}^{\infty} \beta_m \tilde{e}^{-\alpha \beta_m^2 t} \frac{\beta_m^2 + H^2}{L(\beta_m^2 + H^2) + H} \sin \beta_m L \int_0^L \cos \beta_m x' F(x') dx'$$

and β_m 's are the roots of Eq. (2b)

2-6 In a one-dimensional infinite medium, $-\infty < x < \infty$, initially, the region $a < x < b$ is at a constant temperature T_0 , and everywhere outside this region is at zero temperature. Obtain an expression for the temperature distribution $T(x, t)$ in the medium for times $t > 0$.

The mathematical formulation of the problem is given by

$$\begin{aligned}\frac{\partial^2 T}{\partial x^2} &= \frac{1}{\alpha} \frac{\partial T}{\partial t} & \text{in } -\infty < x < \infty, t > 0 \\ T &= T_0 & \text{in } a < x < b \\ T &= 0 & \text{outside this region}\end{aligned} \quad \left. \vphantom{\begin{aligned} \frac{\partial^2 T}{\partial x^2} = \frac{1}{\alpha} \frac{\partial T}{\partial t} \\ T = T_0 \\ T = 0 \end{aligned}} \right\} \text{ for } t = 0$$

The solution of this problem for the case of more general initial condition (i.e. $T = F(x)$ for $t = 0$) is given by Eq. (2-70).

To obtain solution for this problem we set $F(x) \equiv T_0$ and change the limits of integration to $a \leq x' \leq b$, that is

$$T(x, t) = \frac{T_0}{(4\pi\alpha t)^{1/2}} \int_{-\infty}^{\infty} \exp\left[-\frac{(x-x')^2}{4\alpha t}\right] dx' \quad (1)$$

To perform this integral, a new variable is defined as

$$\eta = \frac{x-x'}{\sqrt{4\alpha t}} \quad \text{or} \quad dx' = -\sqrt{4\alpha t} d\eta \quad (2)$$

Then Eq. (1) becomes

$$\begin{aligned}T(x, t) &= \frac{T_0}{2} \left[\frac{2}{\sqrt{\pi}} \int_0^{\frac{x-a}{\sqrt{4\alpha t}}} e^{-\eta^2} d\eta - \frac{2}{\sqrt{\pi}} \int_0^{\frac{x-b}{\sqrt{4\alpha t}}} e^{-\eta^2} d\eta \right] \\ &= \frac{T_0}{2} \left[\operatorname{erf}\left(\frac{x-a}{\sqrt{4\alpha t}}\right) + \operatorname{erf}\left(\frac{b-x}{\sqrt{4\alpha t}}\right) \right]\end{aligned}$$

$$\frac{T(x, t)}{T_0} = \frac{1}{2} \left[\operatorname{erf}\left(\frac{x-a}{\sqrt{4\alpha t}}\right) + \operatorname{erf}\left(\frac{b-x}{\sqrt{4\alpha t}}\right) \right] \quad (3)$$

2-15 Obtain an expression for the steady-state temperature distribution $T(x, y)$ in a rectangular region $0 \leq x \leq a, 0 \leq y \leq b$ for the following boundary conditions: the boundary at $x = 0$ is kept insulated, the boundary at $y = 0$ is kept at a temperature $f(x)$ and the boundaries at $x = a$ and $y = b$ dissipate heat by convection into an environment at zero temperature. Assume the heat transfer coefficient to be the same for both boundaries.

The mathematical formulation of the problem is

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} = 0 \quad \text{in } 0 < x < a, 0 < y < b$$

$$\frac{\partial T}{\partial x} = 0 \quad \text{at } x = 0$$

$$\frac{\partial T}{\partial x} + HT = 0 \quad \text{at } x = a$$

$$T = f(x) \quad \text{at } y = 0$$

$$\frac{\partial T}{\partial y} + HT = 0 \quad \text{at } y = b$$

The separated problems are taken as

$$\left. \begin{aligned} \frac{d^2 X}{dx^2} + \beta^2 X &= 0 \quad \text{in } 0 < x < a \\ \frac{dX}{dx} &= 0 \quad \text{at } x = 0 \\ \frac{dX}{dx} + HX &= 0 \quad \text{at } x = a \end{aligned} \right\} \textcircled{1} \quad \left. \begin{aligned} \frac{d^2 Y}{dy^2} - \beta^2 Y &= 0 \quad \text{in } 0 < y < b \\ \frac{dY}{dy} + HY &= 0 \quad \text{at } y = b \end{aligned} \right\} \textcircled{2}$$

The eigenfunctions, eigencondition and the normalization integral of the problem for the x-separation are obtained from Table 2-2, case #4 by replacing L by " a " and H_2 by H . We find

$$X(\beta_m, x) = \cos \beta_m x \quad \frac{1}{N(\beta_m)} = 2 \frac{\beta_m^2 + H^2}{a(\beta_m^2 + H^2) + H}$$

and β_m 's are the positive roots of $\beta_m \tan \beta_m a = H$.

The solution for the $Y(\beta_m, y)$ function satisfying the boundary condition at $y = b$ of the problem $\textcircled{2}$ is taken as

$$Y(\beta_m, y) = \beta_m \cosh \beta_m (b-y) + H \sinh \beta_m (b-y)$$

The complete solution of the original problem is constructed as

$$T(x, y) = \sum_{m=1}^{\infty} C_m \cos \beta_m x \left[\beta_m \cosh \beta_m (b-y) + H \sinh \beta_m (b-y) \right]$$

This solution is now constrained to satisfy the boundary condition at $y = 0$ of the original problem, i.e. $T = f(x)$ at $y = 0$. We find

$$f(x) = \sum_{m=1}^{\infty} C_m \cos \beta_m x \left[\beta_m \cosh \beta_m b + H \sinh \beta_m b \right]$$

where the unknown coefficients C_m are determined by utilizing the orthogonality of the functions " $\cos \beta_m x$ ".

That is, by operating on both sides of this equation by the operator $\int_0^a \cos \beta_m x \, dx$ and utilizing the orthogonality of $\cos \beta_m x$ functions, C_m is determined as

$$C_m = \frac{1}{N(\beta_m) [\beta_m \cosh \beta_m b + H \sinh \beta_m b]} \int_0^a \cos \beta_m x' f(x') \, dx'$$

Introducing this value of C_m to the above solution, we obtain the solution of the problem as

$$T(x, y) = \sum_{m=1}^{\infty} \frac{\cos \beta_m x \left[\beta_m \cosh \beta_m (b-y) + H \sinh \beta_m (b-y) \right]}{N(\beta_m) [\beta_m \cosh \beta_m b + H \sinh \beta_m b]} \int_0^a \cos \beta_m x' f(x') \, dx'$$

where $\frac{1}{N(\beta_m)} = 2 \frac{\beta_m^2 + H^2}{a(\beta_m^2 + H^2) + H}$ and β_m 's are the roots of $\beta_m \tan \beta_m a = H$.